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In [ ]: import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
from IPython.display import display

RANDOM_STATE = 42
sns.set_theme(style="whitegrid")
plt.rcParams.update({"figure.dpi": 120})

DATA_DIR = Path("../data")
if not DATA_DIR.exists():
    DATA_DIR = Path("Durga/Lab/MLDL/data")

pd.set_option("display.max_columns", 120)

import gzip
import struct

def load_mnist_images(path, limit=None):
    with gzip.open(path, "rb") as fh:
        magic, count, rows, cols = struct.unpack(">IIII", fh.read(16))
        if magic != 2051:
            raise ValueError(f"Unexpected image magic number: {magic}")
        if limit is None:
            limit = count
        data = np.frombuffer(fh.read(rows * cols * limit), dtype=np.uint8)
        return data.reshape(limit, rows, cols).astype("float32") / 255.0

def load_mnist_labels(path, limit=None):
    with gzip.open(path, "rb") as fh:
        magic, count = struct.unpack(">II", fh.read(8))
        if magic != 2049:
            raise ValueError(f"Unexpected label magic number: {magic}")
        if limit is None:
            limit = count
        return np.frombuffer(fh.read(limit), dtype=np.uint8)

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In [2]: import tensorflow as tf
from sklearn.metrics import ConfusionMatrixDisplay, classification_report

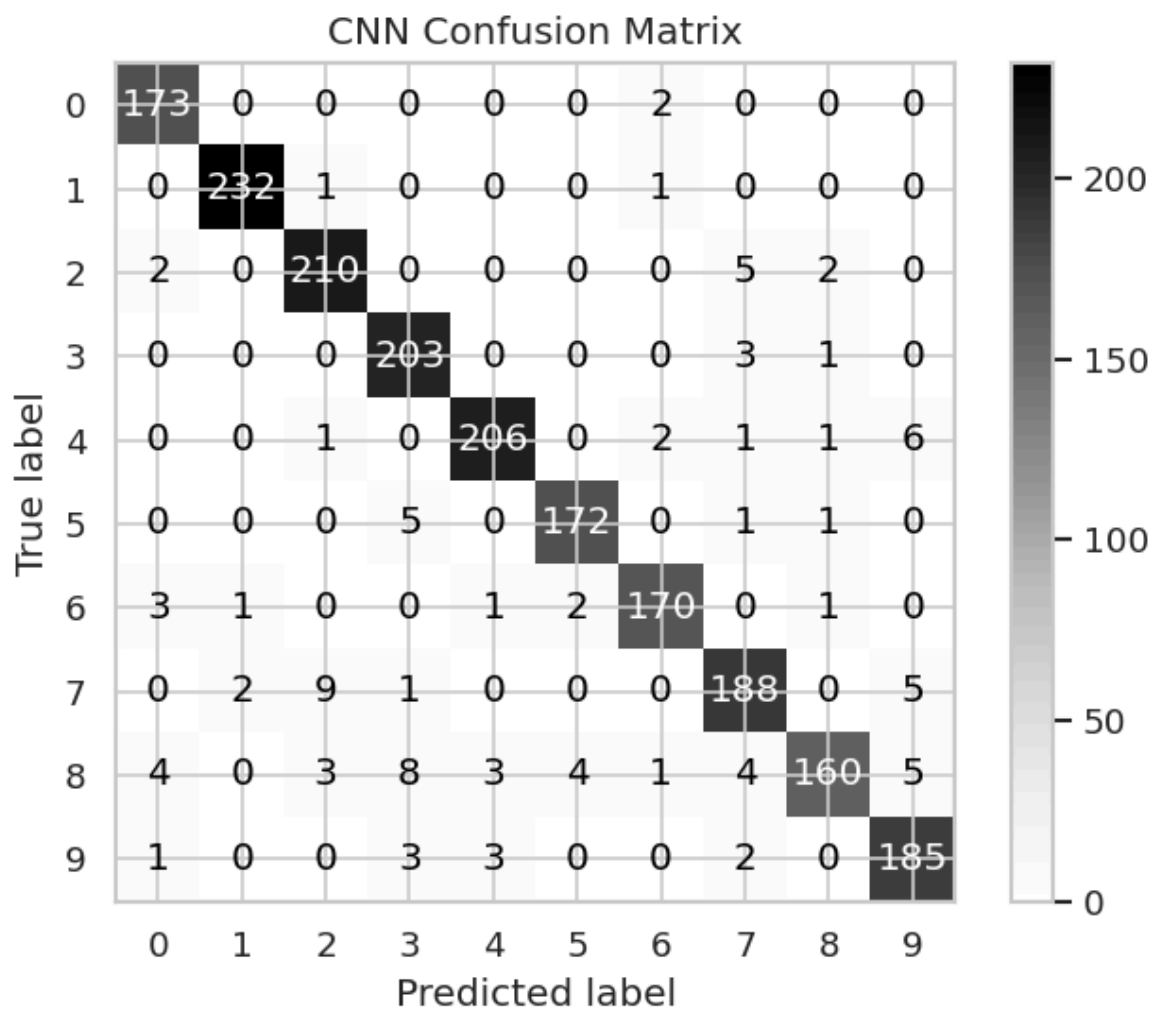
tf.random.set_seed(RANDOM_STATE)

train_images = load_mnist_images(DATA_DIR / "mnist/train-images-idx3-ubyte")
train_labels = load_mnist_labels(DATA_DIR / "mnist/train-labels-idx1-ubyte")
test_images = load_mnist_images(DATA_DIR / "mnist/t10k-images-idx3-ubyte")
test_labels = load_mnist_labels(DATA_DIR / "mnist/t10k-labels-idx1-ubyte")
print(train_images.shape, test_images.shape)

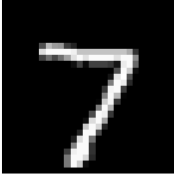
```


[illegible]

	precision	recall	f1-score	support
0	0.95	0.99	0.97	175
1	0.99	0.99	0.99	234
2	0.94	0.96	0.95	219
3	0.92	0.98	0.95	207
4	0.97	0.95	0.96	217
5	0.97	0.96	0.96	179
6	0.97	0.96	0.96	178
7	0.92	0.92	0.92	205
8	0.96	0.83	0.89	192
9	0.92	0.95	0.94	194
accuracy			0.95	2000
macro avg	0.95	0.95	0.95	2000
weighted avg	0.95	0.95	0.95	2000



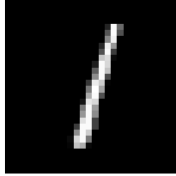
true 7 / pred 7



true 2 / pred 2



true 1 / pred 1



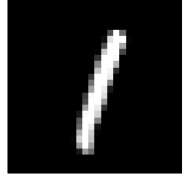
true 0 / pred 0



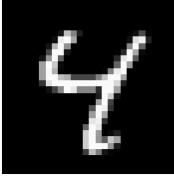
true 4 / pred 4



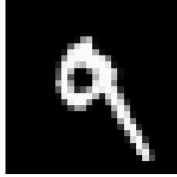
true 1 / pred 1



true 4 / pred 4



true 9 / pred 9



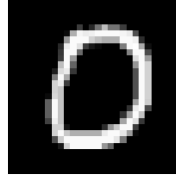
true 5 / pred 5



true 9 / pred 9



true 0 / pred 0



true 6 / pred 6

